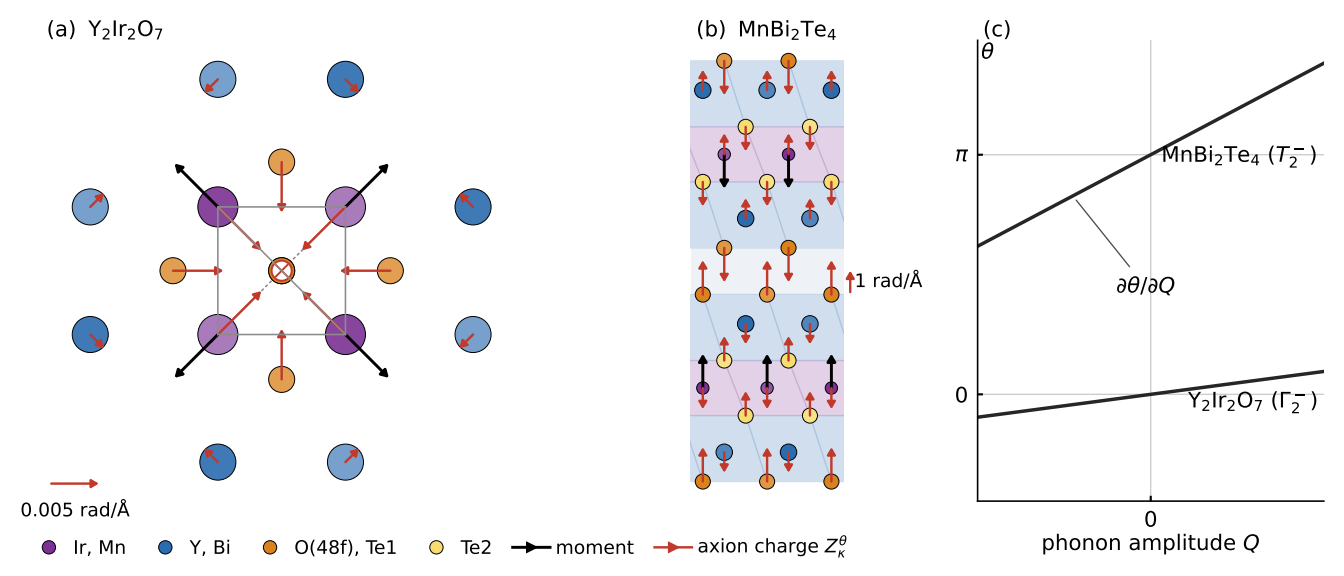
(a) $Y_2Ir_2O_7$
(b) $MnBi_2Te_4$
(c)
θ
π
0
$MnBi_2Te_4$ (T_2^-)
$\partial\theta/\partial Q$
$Y_2Ir_2O_7$ (Γ_2^-)
0
phonon amplitude Q
1 rad/Å
0.005 rad/Å
Ir, Mn
Y, Bi
O(48f), Te1
Te2
moment
axion charge Z_κ^θ